

Actionable Business Architecture

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Abstract—Interviews held with over a thousand CEOs in 2010 and CIOs in 2011 show that organizations are expecting change and are being challenged with complexity. As CEOs and CIOs have started strategically aligning their thinking around change, future challenges, growth and complexity, organizations must now focus on driving new and better business value from the enterprise's portfolio of technology to remain successful. Technology designs and architectures need to realistically identify and represent the business's capabilities and their relationships to operations and technology.

This paper presents a new concept called Actionable Business Architecture. Actionable Business Architecture builds beyond the state-of-the-art in the field of Business Architecture by providing a new perspective on the subject of managing the confluence of three key business models of strategy, operations and technology. It consumes the science and practice of strategy management, business process and case management, service orientation and industry enterprise models as the means to make the business architecture simplified and focused to support transformation.

Keywords: *Business Architecture; Actionable Business Architecture; Business - IT convergence; Operations Strategy; Enterprise Architecture*

I. INTRODUCTION

The growing economic uncertainty and its widening impact create a need for businesses, government organizations and non-profits to seek ways to accomplish their business objectives more effectively and faster than ever before [1]. Successful transformation requires more relevant and focused designs which in turn require having an actionable business architecture as a control point for business relevancy assurance in the design paradigm.

This paper discusses what makes Business Architecture actionable from three specific perspectives: strategy and transformation (S&T), business process management (BPM) and service oriented architecture (SOA). The ability to act leveraging the business architecture is critical so that the architecture is viewed not as a drain on projects, but rather as a significant cost saver, a critical enabler for business differentiation and revenue generation. This is truer in today's business climate more than ever before. Organizations are seeking out realistic approaches that enable them to develop and maintain a strong and actionable Business Architecture.

Business Architecture defines and manages the valued relationships and interactions among the strategy, operating and IT models of a business. Business Architecture is, as reflected in Fig. 1, at the confluence of these three models.

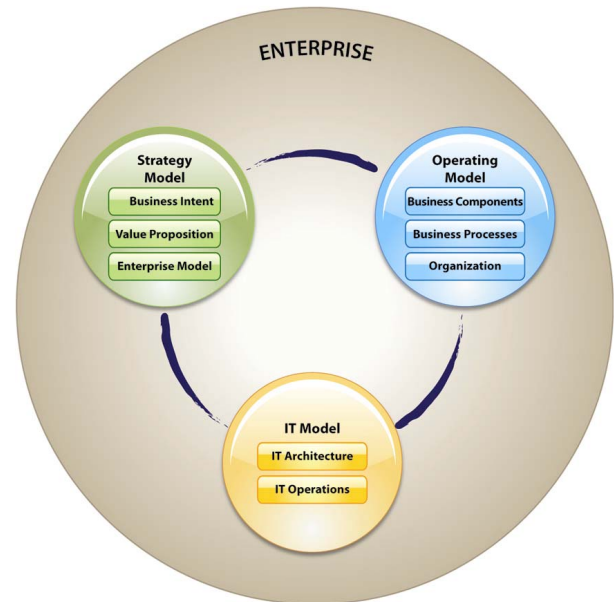


Figure 1. Positioning of Business Architecture within an Enterprise and its key dimensions

Therefore, Business Architecture, at its core, has elements and connections to ensure that the strategic intent of the business is effectively executed - at the operational level through its business processes, and supported and enabled by the various IT services and systems.

In order to enumerate the various dimensions that represent the Business Architecture of an enterprise, it is important to understand what is contained within each of the strategy, operating and IT models. Fig. 1 shows some of the typical dimensions that compose a Business Architecture.

What makes Business Architecture actionable is that it's realized through prescriptive approaches guided by robust techniques. It must also be formally represented through a detailed set of rich artifacts. Each of the three models, namely strategy, operating and IT, are realized and managed through the following respective contexts:

- The strategy model is best realized through a strategy and transformation context that explores the details of the strategy and captures the specifics through models and codified artifacts that drive competitive advantage

in an organization to makes business performance sustainable.

- The operating model is best realized through business process management (BPM) which is a well known context [2] that expresses the details of the execution and operation of the organization.
- The IT model can be realized through a variety of contexts. Service orientation as a concept is most applicable across all three of the models and hence service oriented architecture (SOA) is chosen as the context to discuss the realization of the IT model.

The intersections of the contexts yield Actionable Business Architecture (Fig. 2) since it is codified through a rich set of artifacts and with the corresponding prescriptive approaches.

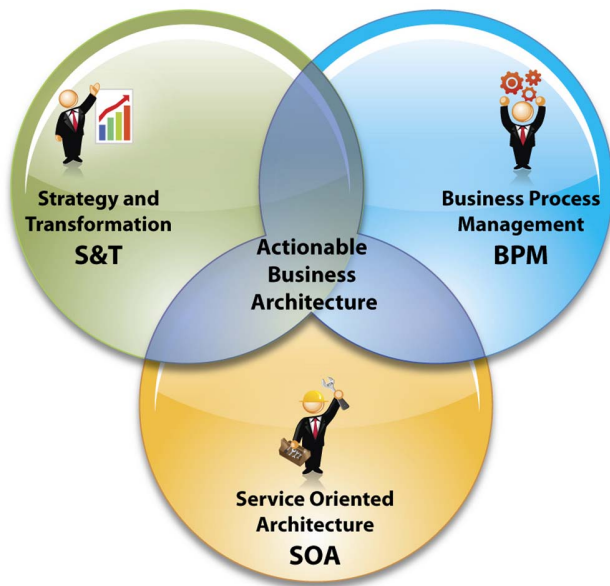


Figure 2. Actionable Business Architecture is the intersections of the contexts

II. BUSINESS ARCHITECTURE AND ENTERPRISE ARCHITECTURE

While it is true that in most industries enterprise architecture [3] has found acceptance as a vehicle to align business and IT [4], the perception of many business leaders is that enterprise architecture serves and resides only within the IT unit of the organization. This trend varies with the geography of the organization, the dynamics of the industry or industry segment, and the degree of maturity of an organization's operations. The perception has evolved over time as stakeholders began to realize that within the context of business and IT alignment, an enterprise architecture must explicitly include the business aspects of the enterprise as well as the underlying IT support.

Incorporating business architecture as one of the domains that comprise enterprise architecture is in direct support of a

definition of enterprise architecture as a discipline that defines and maintains the architecture models, governance and transition initiatives needed to effectively coordinate semi-autonomous groups towards common business and/or IT goals. This definition highlights the strong relationship between enterprise architecture and its outcome in supporting business goals and objectives; hence the need for better articulation of the business objectives, goals, and processes that can be best captured and articulated by business architecture as defined earlier.

Others have also defined enterprise architecture in similar terms all emphasizing the value in achieving alignment of business and IT. For example, Gartner defined enterprise architecture in several white papers as the process that translates business vision and strategy into effective enterprise change by creating, communicating and improving the principles and models that describe the enterprise's future state and enable its evolution [5].

Though nearly all these definitions mentioned the business aspect of enterprise architecture, historically the lack of strong and formal representation of the business aspects in enterprise architecture engagements rendered the goal of achieving better alignment between business and IT ineffective. Recognizing this deficiency and to ensure proper inclusion and representation of the business, most enterprise architecture frameworks and methods currently in use consider Business Architecture as an essential domain of enterprise architecture [6].

The development of Business Architecture is a principal activity that needs to be undertaken by enterprise architecture unless business architecture artifacts have already been developed as part of other organizational initiatives, such as enterprise planning, strategic business planning or business process re-engineering. In this case, these artifacts become part of the resulting enterprise architecture structure and become an essential input to the development of the rest of the enterprise architecture domains: information systems architecture, technology architecture, and gap analysis and transition planning. For example, target enterprise capabilities and the resources needed to support these capabilities—business services, business users' roles, business processes, and locations where business is being conducted — are all artifacts of business architecture that are needed as input for identifying the other aspects of enterprise architecture. This includes both the functional and operational views of information systems and technology architecture.

However, considering business architecture as an integral part of enterprise architecture doesn't mean that business architecture must be used exclusively in the context of enterprise architecture, or for that matter, produced entirely within the enterprise architecture discipline. As an example, business architecture can be used outside the scope of enterprise architecture, if an enterprise is embarked on a business process reengineering and optimization as part of strategy and transformation initiative; implementation and execution of business processes as part of BPM initiative; or as input to SOA adoption program. Nevertheless, it is important to

keep in mind that enterprise architecture will not be realized unless its activities include business architecture.

For any enterprise architecture program to be effective, development of business architecture artifacts should be included as an integral part of the program activities. In that sense, and with the understanding that business architecture can be used outside the scope of enterprise architecture, we regard enterprise architecture as an umbrella under which both Business Architecture as well as IT Architecture is integrated with continuous feedback between them to achieve the illusive business and IT Alignment that has been sought for the past several decades.

III. ACTIONABLE BUSINESS ARCHITECTURE

The following four keys define the approach to Actionable Business Architecture presented in this paper: models, metrics, methods and tooling.

Models are reusable assets, industry leading practices and industry standards or guidelines that capture knowledge about the industry and the enterprise. They can be leveraged successfully to accelerate the creation of Actionable Business Architecture. These models should be available and leveraged during Strategy and Transformation, BPM and SOA initiatives.

Methods are techniques that weave through the various contexts using proven methods in the IBM® Component Business Model™, Business Process Management (BPM) [7] and Service-Oriented Modeling and Architecture (SOMA). Methods should also describe the how-to of execution while enabling further integration into an enterprise architecture [8] context method.

Metrics are key performance, agility and risk indicators, service levels and industry measures that provide guidance on what to measure in an organization to ensure that business goals and objectives are being met during execution and operations of the business, or through periods of transition.

Tooling helps automate the use of models, methods and metrics. Tooling supports the creation of models at the appropriate stages, brings the right metrics at various levels, and facilitates the use of relevant best-practices.

Fig. 3 illustrates the four keys as the core of the approach to Actionable Business Architecture presented in this paper. It is imperative that the four keys interrelate and operate in concert to maximize the prescriptive value and efficiency of an Actionable Business Architecture.

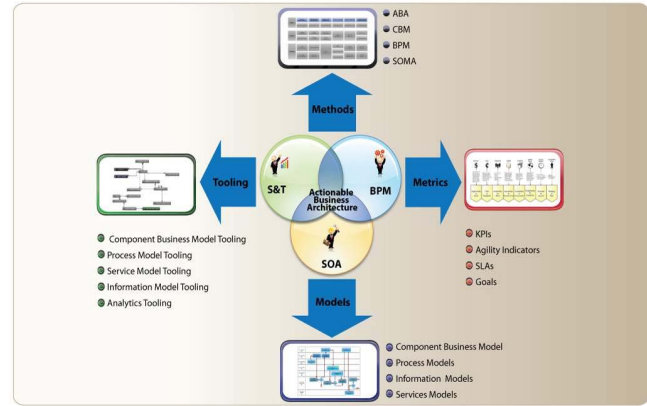


Figure 3. The Four Keys of IBM's approach to Actionable Business Architecture

A. Models

Industry business architecture models (IBAM) contain both representations as well as content and combine industry-specific knowledge with suitable structure. The architecture organizes the content according to the specific industry business dimensions and their interrelationships. IBAMs are critical to the approach for Actionable Business Architecture, providing the following characteristics and strengths:

- Embed industry variation and specialization
- Conform to prescriptive methods that are easily consumable by corresponding tooling
- Promote the adoption of industry and cross-industry standards whenever possible
- Capture and codify the experiences gained from numerous projects, which enable the creation of models beyond the common boundaries of an industry

IBAMs typically contain artifacts representing Strategy and Transformation, Business Process Management and Service-Oriented Architecture. Specifically, the constituent elements are:

- Component Business Models
- Business process models
- Business and technical services
- Information/data models

These elements of an IBAM have some important and traceable interrelationships. For example, business processes relate to the business components by using the capabilities, activities and resources that define components; information models describe data that is used and transformed by business processes; services instrument the capabilities necessary to implement the processes, and so on. While some of these concepts are rather known and belong to a wider scope of standards, others are novel for Business Architecture, such as Component Business Modeling (see Fig. 4 for a two-dimensional view of the model). Understanding and leveraging

these relationships across the modeling domains is critical for any solid business architecture.



Figure 4. Component Business Model representation of the enterprise.

B. Methods

The ultimate goal of any business transformation is to create value. To achieve this goal, it is essential to focus on value creation, measurement and sustainability through a holistic delivery approach of business solutions spanning business strategy formulation, execution and implementation. Actionable Business Architecture achieves this goal and creates value by closing the gaps between strategy, operations and IT implementation.

Using this method, enterprises can maximize their return on investment from business or IT transformations by tackling them holistically through an end-to-end approach, rather than concentrating on isolated initiatives or point solutions in strategy, operations or IT. This method (see Fig. 5) has a number of phases, some of them optional, which take a series of transformation initiatives to be rationalized and instrumented by following a SOA solution approach if needed.

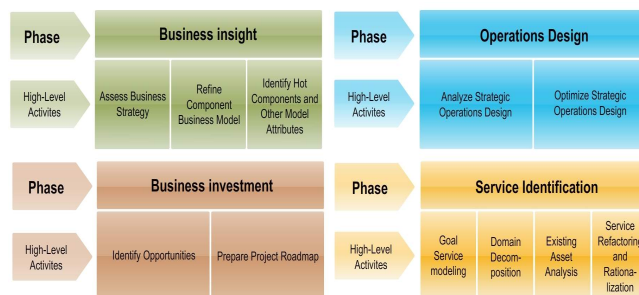


Figure 5. Actionable Business Architecture method

Such an approach creates a continuous improvement environment for measuring and validating anticipated business outcomes. More importantly, Actionable Business Architecture links these outcomes to business value by having the ability to trace them across strategy, operations and IT. Some of the tangible business outcomes of using ABA methodology can be summarized as follows:

- Shift business model to focus on revenue and innovation by quickly combining and evolving

reusable components, creating new ways to capture business opportunities.

- Reduce risk with predictable results and higher quality.
- Decrease labor costs and enhance ability to reuse and transfer skills between projects, business units, or service lines across the organization.
- Drive sustainable, consistent, and effective reuse of assets and best practices by ensuring that methods are supported by tooling.
- Increase time to market and customer success by leveraging industry leading best practices.

Even when effectively practicing strategy formulation, business process management or service-oriented architecture methodologies in isolation, organizations may still experience suboptimal business outcomes. Common scenarios that are manifested when the three practices are used in isolation are:

- Business transformation is underway but it is unclear or uncertain as to whether the IT projects that have been launched are consistent with the business goals
- Business and IT transformation is underway where the business strategy is being defined or refined and there is a requirement to redefine a supporting IT direction, architecture or strategy
- Technology oriented or IT strategy project launched, but it becomes evident that the necessary business linkage or commitment is not sufficient

As depicted in Fig. 6, the approach to Actionable Business Architecture includes and connects a discrete set of methods spanning and integrating strategy, operations and IT. These methods address the preferred realizations defining Actionable Business Architecture, namely, Strategy and Transformation using CBM Method [9], Business Process Management using BPM Method [10] and Service Oriented Architecture using SOMA Method [11, 12]). Moreover, the Actionable Business Architecture method introduced in this section specifically provides a deep integration across these individual methods and builds a traceable path from business performance priorities to IT implementation.

In general, an Actionable Business Architecture should accomplish the following:

- Support the realization and relevant interlocking among key models
- Enable a more effective business focus around creation, usage and subsequent reuse of enterprise-wide assets
- Prescribe the correct variety of techniques, artifacts, information and analytics needed to achieve business and IT convergence
- Identify the active roles needed to invest in the life cycle processes

- Ensure the appropriate level of organizational participation and control necessary in a role-based fashion

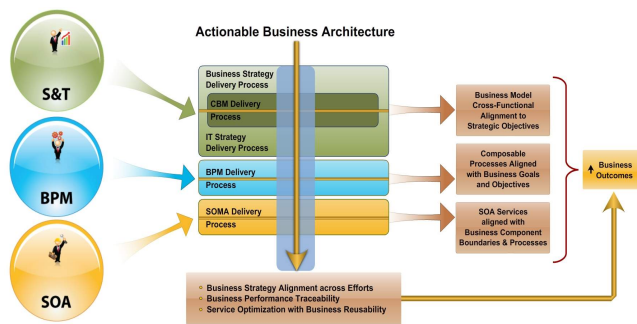


Figure 6. Actionable Business Architecture method and its relationship to CBM, BPM and SOMA methods

C. Metrics

An enterprise must adequately measure performance to drive improved business results. Actionable Business Architecture enables a powerful use of metrics, intelligently integrated into models for ease of use and fact-based decision making. Metrics set the business performance expectation for the organization. They equip Business Architecture with feedback to understand the health and agility of the business. The main aspects of managing performance in an organization includes: *The Anatomy of a Metric*, *Managing the Lifecycle of Metrics*, *Metrics as a Source of Traceability* and *Applying Benchmarks*.

Organizations must determine which metrics are critical to managing the business—these priority metrics are called Key Performance Indicators. While every KPI is a metric, not all metrics are KPIs. The Benchmarking Program® in IBM has defined and captured benchmarking data for over 1,000 metrics. Clearly an organization cannot and should not actively manage hundreds of metrics. A smaller set of integrated metrics ensures that the organization works towards its strategic priorities. The IBM Benchmark Program¹ leverages APQC's Open Standards Benchmarking Collaborative (OSBC)² that has data from over 18,000 organization entities worldwide and which recognizes the value of scale and consistency to drive quality.

Measurements, by themselves, do not reveal the action required to improve the business. Insight from assessment leads to actions that improve performance and efficiencies. The more the metrics approach is performed, the more it becomes an integral fabric of the organization and a core part of its culture.

Regardless of how an organization starts using business architecture, organizations can trace their performance, down to process and up to strategies, through Actionable Business

Architecture, which values and leverages metrics. In IBAM, metrics are embedded into models to address business performance according to the operating entity being modeled. Furthermore, it is important to support performance management with a collection of tooling for Actionable Business Architecture, business intelligence and analytics that helps organization track and monitor metrics. During the IBM 2010 CFO Study [13], one CFO proclaimed, "If you can't believe the data, how will you believe the analytics?" Indeed the quality of the data being measured is not to be underestimated. Metrics produce a sort of mind map that should be understood. Fig. 7 is an illustration that highlights the multiple integration points metrics have on one another. This visual representation of these relationships can help some business leaders appreciate the importance of defining and managing the right set of measures for their organization. This integration brings analytics to be reconciled with the rest of the architecture of an enterprise. This subject goes beyond the scope of this paper.

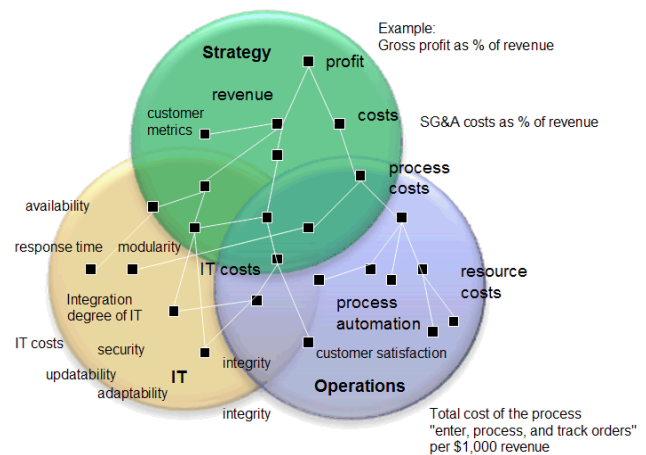


Figure 7. The relationship among metrics across the domains of Actionable Business Architecture

D. Tooling

Tooling embeds the necessary best-practices, provides adequate guidance for the entire life cycle and supports ongoing collaboration across a wide variety of roles in the organization. These integrated capabilities support a successful deployment and management of Actionable Business Architecture.

Suitable tooling needs to address Strategy and Transformation, BPM and SOA individually. But more importantly, it needs to provide the necessary interlocks across strategy, operations and IT to enforce those conditions that make their integration viable. Tooling that addresses Strategy and Transformation, BPM and SOA in isolation typically define and manage their own entities and relationships in a siloed approach. For example, business capabilities defined and managed by a Strategy and Transformation tool may not be easily kept in alignment with resources that are consumed or

¹ IBM Institute for Business Value Benchmark Program, contact: global.benchmarking@us.ibm.com

² American Process Quality Collaborative (APQC), www.apqc.org.

produced by business processes in a BPM tool. Similarly, goals involved in an SOA tool may not be readily integrated with the world of BPM or strategy. This means changes in such goals dictated by new operational or strategic conditions will not be interlocked holistically.

An effective approach to Business Architecture should include a unified set of tooling that realizes the above goals. The tooling used in Actionable Business Architecture comes from IBM Software Group, IBM Global Business Services and IBM Research. For example, the Component Business Modeling Tool supports Strategy and Transformation at both corporate and business strategy levels. The CBM Tool (see Fig. 8) is an asset that provides the capabilities necessary to model organizations by using CBM and supports the deployment of the CBM Method. Furthermore, it provides a tight integration with the other tooling necessary to support Actionable Business Architecture.

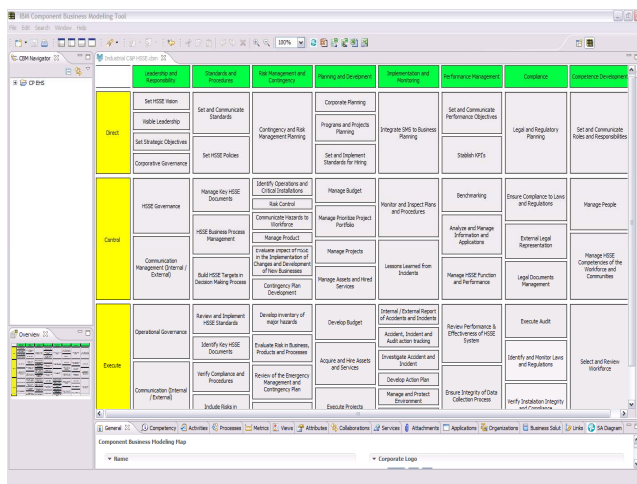


Figure 8. Component Business Modeling Tool showing a two dimensional representation of the business components defining the Actionable Business Architecture of the organization.

Another differentiated asset from the SOMA-Modeling Environment (SOMA-ME) supports the SOMA Method by providing a tight integration with the strategy and operating models. For example, SOMA-ME works on the business goals and prioritized business processes defined by the CBM Tool. In addition, other tooling from IBM Software Group for Actionable Business Architecture includes integrated functionality for BPM, information modeling, enterprise architecture and business performance management.

- Protecting investments: adequately tooling implements a flexible and extensible software architecture that can help integrate with existing platforms and develop customization for specific industries.
- User tooling adoption: Because many organizations do not have modeling skills within their company, suitable tooling should come with field adoption patterns that

have been proven both acceptable and efficient for the impacted modeling and analysis user communities.

- Analytics: The ability to take initial, updated and interlocked models and use them with tooling to perform "what if" innovation scenarios for processes and services.
- Collaboration: The ability to provide a facility to expose the models in a fashion that allows their publication and communication, as well as to foster collaboration with the broader organization.
- Enterprise Architecture Supportive: Organizations that actively support enterprise architecture activities are able to utilize the Actionable Business Architecture tool-suite seamlessly in the enterprise architecture context.
- Costing: The ability to persist and augment models with cost factors, such as resource costs, and system/component service costs, so that the financial impact of identified transformation and innovation initiatives can be assessed.

Ultimately, tooling required for model creation, governance and method deployment are an expense that is recouped in the consistency and accurate speed to market that the right tooling brings to an Actionable Business Architecture journey. But a critical point regarding this expense is that tooling is used to support both individual methods and models, and end-to-end context preservation. This is so that the models and methods grow in depth and richness around the business context. IBM has invested significantly in creating and improving its method based tool-suite so that it is consumable by a dedicated user community and can still leverage a wide variety of existing user skills and previous IT investments

IV. CONCLUSION

Embracing the tenets and prescription around adoption of an Actionable Business Architecture will drive continuous improvement. It will also help an organization maintain the higher valued relationships within an enterprise in a more usable context to fuel sustainable growth.

The authors believe, and practical field experience has demonstrated, that Actionable Business Architecture instruments business and IT convergence using an interlocked set of models, methods, metrics and tooling. Actionable Business Architecture enables a delivery more rapidly, accurately and efficiently to focus organizational efforts towards a desired business context.

The review of each of the four keys, models, methods, metrics, and tooling, has shown that they individually have a unique and broader context. The dependency and relationship of each to one or more of the others drives the conclusion that when executed in isolation each key could still create value, but only in a smaller increment limited to their context. Actionable Business Architecture requires accomplishing only what is necessary from within each key, but always in concert with the others to ensure that maximum cumulative value is derived from every transformative effort.

Actionable Business Architecture generates a framework for the reliable and repeatable delivery of positive business results through definition and documentation of valued relationships with visibility of context, convergence of purpose, and traceability across strategic efforts, both immediately and over time. Starting with aligned models and metrics delivered with the controlled means and manner of methods and tooling, optimizes their collaborative usage to ensure convergence, cohesive focus and differentiated acceleration of change and transformation. This value accumulation guides how the Actionable Business Architecture can be greater than the sum of its parts and how business agility and flexibility can be more readily achieved and repeated to deliver more positive business outcomes.

Initial field practice with Actionable Business Architecture has shown it handles the details around the complexity of associating and defining higher valued relationships by containing them through utilization of four keys: to manage the four representative contexts, across eight core dimensions, within three organizationally significant models of strategy, operating and IT. The details for how to get started and drive towards life cycle management will be introduced in the presentation of this paper at the workshop and will also be documented in another whitepaper.

Ultimately, Actionable Business Architecture is designed to provide a systematic, right-sized and progressive approach to breaking down a complex enterprise to produce cost savings and business differentiation through convergence of purpose and consistent execution across the organization. There is more information available regarding Actionable Business Architecture and IBM's approach in two whitepapers [14, 15].

Important to this discourse is the concept of introducing the choices and patterns related to initiating an Actionable Business Architecture as a means towards achieving a steady state of business agility and converged flexibility through leveraging of a vibrant, living business architecture.

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Additionally the authors would like to acknowledge the above persons for their related domain work and expertise over the last decade. They have advanced the respective state of the art for their respective fields of practice. Their foundational work enhanced the domain of business architecture and ultimately culminated in the definition and refinement of what is called Actionable Business Architecture as we present it today.

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